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Paper 1: Context Engineering for AI-Assisted Pharmacometrics: A Practical Tutorial

Bibliographic Information

- **Title:** Context Engineering for AI-Assisted Pharmacometrics: A Practical Tutorial
- **Authors:** Pritchard-Bell A, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026
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Paper Type: AI_ML

Quick Take

This tutorial presents a context engineering framework for LLM-assisted pharmacometric analysis, where each workflow step runs as a self-contained task in a fresh LLM instance with embedded domain rules, verification criteria, and file-based state transfer. Using a synthetic PK/PD scenario, the authors show that rule specificity predicts output consistency, compressing dose recommendations from an 8-fold to a 1.5-fold spread across six cross-platform runs.

Executive Summary

The paper addresses the reliability gap in LLM-assisted pharmacometrics by introducing a task library architecture in which domain-specific rules, verification criteria, and worked examples are embedded directly into each task definition (JSON schema with 8 fields). Each task runs in a fresh LLM instance with clean context, with information passed between tasks through shared workspace files, avoiding context degradation and error propagation. The framework was evaluated on a synthetic vorlizumab (mAb) PK/PD scenario across two agentic platforms (Claude Code/Opus 4.6 and OpenAI Codex/GPT-5.3, 3 replicates each), demonstrating that explicit rules drive convergence on core analytical decisions (2-compartment model, FOCE-I, allometric scaling, M3 BLQ handling) while remaining divergences map to genuine methodological latitude requiring human adjudication. The divergence-driven refinement method—using run-to-run nondeterminism as a diagnostic for locating underspecified rules—is a notable methodological contribution.

Scientific Context & Motivation

LLMs can execute pharmacometric workflows but make consequential domain-specific errors when task instructions lack adequate details; prior evaluations found LLMs required expert correction for reliable outputs. The paper argues that context engineering—controlling what information reaches the LLM at each decision point—is the primary determinant of reliability, rather than raw model capability. Pharmacometrics is well-suited to this approach due to separable tasks (data prep, structural modeling, covariate analysis, simulation, documentation),

file-based outputs, enumerable domain rules (FOCE-I, allometric scaling, $\Delta OFV > 3.84$), and known-answer testing via synthetic scenarios with ground-truth parameters.

✂ Methodological Snapshot

The framework has three independently varying layers: task definitions (JSON with 8 fields: task_id, category, description, produces, verify, rules, depends_on, passes), an outer loop (deterministic shell script managing agent lifecycle via sentinel files ANALYSIS_COMPLETE.txt and ESCALATE.txt), and an inner agent (any LLM with tool-use capabilities). Each task runs in a fresh LLM instance with context loaded in layers: universal protocol prompt, progress log (accumulated findings), task-specific definition, study_context.md (project-level domain knowledge), and on-demand examples/upstream outputs. Information passes between tasks through workspace files, not conversation memory. Refinement uses two mechanisms: expert review (mapping disagreements to missing/underspecified rules) and divergence-driven replication (running the same library across platforms to locate rule gaps where runs diverge).

🤖 Task Framing

The task is orchestrating multi-step pharmacometric workflows (data preparation, exploratory analysis, structural model development, covariate analysis, simulation, documentation) using LLM agents. It is framed as a context engineering problem: providing each decision point with the right domain knowledge, study details, and upstream results without context pollution. The evaluation uses a synthetic known-answer benchmark (vorlizumab mAb PK/PD) to test whether rule specificity predicts output consistency across platforms.

Dataset Description

Synthetic vorlizumab scenario (a monoclonal antibody) with known ground-truth PK/PD parameters. ADaM-compliant synthetic datasets (ADSL, ADEX, ADPC, ADLB, ADEFF, ADAE, ADADA) for two studies, including SC-dominant sparse PK sampling (which produced high η -shrinkage on V_1). Six runs total: 3 replicates on Claude Code/Opus 4.6 and 3 on OpenAI Codex/GPT-5.3, with model versions pinned. No train/test split—this is an agentic workflow evaluation, not supervised learning; the ‘labels’ are the ground-truth PK/PD parameters used for recovery assessment.

Model Architecture

Not a traditional ML model. The ‘model’ is the task library + agentic harness. Task definitions use a JSON schema with 8 fields. Inner agents are Claude Code (driving Opus 4.6) and OpenAI Codex (driving GPT-5.3), both with tool-use capabilities. Pharmacometric tools: mrgsolve/nls.lm for naive pooled structural screening, nlmixr2 for NLME modeling with FOCE-I estimation. The outer loop is a deterministic shell script performing topological sort of the dependency graph, spawning one agent instance at a time sequentially.

Training Details

No model training. Refinement is prompt/context engineering: two iterative cycles of expert review and cross-platform replication. Rule additions were targeted to specific gaps: ‘Estimate F_1 when IV data present’ (added to structural_1cpt_fit, structural_2cpt_fit, base_model), ‘Non-parametric bootstrap of final model’ (parametric enforced for evaluation runtime), and ‘Indirect response as primary PD candidate’ (pkpd_model task). Model versions pinned at evaluation time. Each full run completed in ~4 h at ~\$70 API cost (Claude Code/Opus 4.6).

Evaluation Metrics

Convergence across runs on core analytical decisions (structural model, estimation method, allometric scaling, BLQ handling, error model); parameter recovery vs. ground truth (F_1 : 0.64–0.75 vs. 0.70; EC_{50}/IC_{50} : 0.56–2.80 mg/L vs. 2.50); dose recommendation spread (8-fold $\rightarrow$ 1.5-fold); CL spread reduction (30% $\rightarrow$ 10.5%); AIC/BIC for structural model comparison ($\Delta AIC = 143.6$ favoring 2-compartment). No formal statistical tests; comparisons are descriptive across $N = 6$ runs.

Deployment Caveats

LLM outputs are non-deterministic even at temperature zero (5–15% parameter variation across replicates); users should run multiple replicates and pin model versions. Production deployments require sandboxed environments with file system access limited to the workspace and code execution confined to the R session; defense-in-depth (sandboxing, restricted permissions, review checkpoints, workspace auditing) is recommended. The framework has not been tested in prospective regulatory submissions. Remaining divergence (dose 400–600 mg, disease covariate retention, E-R significance) reflects genuine methodological latitude requiring expert adjudication at human checkpoints. The ‘indirect response’ rule is study-specific and would be the wrong default for other mechanisms.

☒ Key Findings

Rule specificity predicted output consistency: six runs across two platforms converged on core decisions (2-compartment structural model, FOCE-I estimation, allometric scaling with fixed exponents, M3 BLQ handling, combined proportional-additive error) where task definitions were explicit. Three rule gaps were identified and fixed via targeted edits: (1) adding ‘Estimate F_1 when IV data present’ reduced between-platform CL spread from 30% to 10.5% and recovered F_1 (0.64–0.75 vs. truth 0.70); (2) standardizing bootstrap method resolved non-parametric vs. parametric divergence; (3) tightening PD model form to ‘indirect response as primary candidate’ compressed EC_{50}/IC_{50} range and dose recommendations from 8-fold (100–800 mg) to 1.5-fold (400–600 mg monthly). Self-verification reliably catches mechanical failures (non-convergence, implausible SEs) but cannot adjudicate scientific judgment—every divergent PD model passed self-verification—so human review checkpoints (ESCALATE.txt) remain essential. Remaining divergence (dose

400–600 mg, disease covariate retention, E-R significance) reflects genuine methodological latitude, not rule gaps, and is reframed as variation-as-exploration.

Strengths & Limitations

Strengths

- Novel context engineering framework adapted to pharmacometrics, with clean separation of task definitions, outer loop, and inner agent
- Divergence-driven refinement method: cross-platform replication turns run-to-run nondeterminism into a diagnostic for locating underspecified rules
- Traceability and audit trail via progress log, file-based outputs, and human checkpoints (ESCALATE.txt) at phase boundaries
- Reproducible synthetic benchmark with ground-truth PK/PD parameters enabling known-answer testing
- Practical cost/performance data (~4 h, ~\$70 per full run on Claude Code/Opus 4.6)
- Honest treatment of remaining divergence as genuine methodological latitude rather than failure
- Alignment with emerging regulatory expectations for AI in drug development (traceability, audit trails, human oversight)

Limitations (Acknowledged by Authors)

- Small N (3 replicates per platform) on a single synthetic scenario; within-platform variance estimates imprecise
- Harness effects (Claude Code vs. Codex) and LLM capability effects (Opus 4.6 vs. GPT-5.3) are confounded
- Expert review involved only four pharmacometricians from the development team; no external validation
- Framework not tested in prospective regulatory submissions
- LLM outputs non-deterministic even at temperature zero (5–15% parameter variation across replicates)
- IIV on V_1 poorly recovered (84–93% below truth) due to high η -shrinkage in SC-dominant sparse design

Limitations (Expert Review)

- The ‘indirect response as primary candidate’ rule is study-specific (appropriate for suppression-mechanism biomarkers) and not transferable; the authors acknowledge this but the generalizable formulation (‘consistent with mechanism of action’) defers judgment rather than resolving it
- Sequential execution limits scalability; parallel execution of independent branches is mentioned but not implemented or benchmarked
- No formal statistical testing of convergence/divergence across runs; conclusions are qualitative
- Cost/time estimates based on a single platform; cross-platform cost variability not reported

- The framework’s reliance on expert review at checkpoints may not scale to routine high-throughput use
- The ‘timetravel’ rewind capability is described but not demonstrated in the evaluation

Generalizability

The framework is domain-agnostic in principle (any domain with separable tasks, file-based outputs, enumerable rules, and testable verification criteria) but demonstrated only on a single synthetic mAb PK/PD scenario. Applicability to small molecules, pediatric/special populations, NONMEM-centric workflows, PBPK, QSP, and regulatory submissions remains untested. The task library is pharmacometrics-specific, but the architecture (task definitions + outer loop + inner agent) generalizes; the authors note the ‘indirect response’ rule is study-specific and would be the wrong default for direct or effect-compartment mechanisms.

Key Equations

Akaike Information Criterion (AIC)

$$AIC = n \cdot \log(RSS/n) + 2k$$

Used for model comparison in the task library (e.g., 1-compartment vs. 2-compartment structural model selection), where n is the number of observations, RSS is the residual sum of squares, and k is the number of parameters.

Covariate inclusion threshold

$$\Delta OFV > 3.84$$

Domain-specific rule for covariate inclusion in forward selection, corresponding to the chi-square critical value at $p < 0.05$ with 1 degree of freedom; embedded in `covariate_forward` and `covariate_backward` task definitions.

Figures & Tables

- **Figure 1:** Three-layer architecture: task definitions (what to produce, how to verify) decoupled from the outer loop (instantiation strategy) and inner agent (task execution).
 - *Significance:* Establishes the core design principle that task design, orchestration, and agent implementation vary independently, enabling benchmarking across platforms and targeted edits.
- **Figure 2:** Execution flow: the outer loop spawns fresh agent instances using sentinel files (ANALYSIS_COMPLETE.txt, ESCALATE.txt) for control flow, with dependency resolution via the task graph.
 - *Significance:* Illustrates the context isolation mechanism—each task runs in a fresh LLM instance with file-based state transfer—which is central to preventing context degradation.
- **Figure 3:** Context loading strategy: solid boxes (protocol, study_context.md, task definition, progress log) always loaded; dashed boxes (examples, upstream outputs) discoverable and loaded on demand.

- *Significance*: Shows how context is layered to keep each decision point focused, with `study_context.md` providing project-level domain knowledge stated once and available everywhere.
- **Figure 4**: Cross-platform evaluation before (panels A–B) and after (panels C–D) rule fixes, showing F_1 estimation, PD model form, and dose recommendation spread across six runs.
 - *Significance*: Directly demonstrates the central claim that rule specificity predicts output consistency, with dose recommendations compressing from 8-fold to 1.5-fold after targeted rule additions.
- **Table 1**: Task schema: eight fields (`task_id`, `category`, `description`, `produces`, `verify`, `rules`, `depends_on`, `passes`) with their purposes.
 - *Significance*: Defines the minimal structured format for task definitions, the core unit of the framework.
- **Table 2**: Expert feedback change types: task additions (5 review tasks), `study_context.md` enrichment, and example expansion, with implementations.
 - *Significance*: Shows the three mechanisms by which expert review improves the task library, increasing it from 30 to 36 tasks.
- **Table 3**: Selected analytical decisions before and after rule fixes: F_1 estimation, bootstrap method, PD model form, EC_{50}/IC_{50} recovery, and recommended dose.
 - *Significance*: Quantifies the impact of rule additions on output consistency across the six-run evaluation.

Code & Reproducibility Assessment

The complete task library (36 task definitions with domain rules and verification criteria), workflow protocol, outer loop pseudocode, and implementation guide are provided in Supporting Information S1. The synthetic vorlizumab scenario is fully specified in S5 with ground-truth PK/PD parameters, and ADaM-compliant synthetic datasets (ADSL, ADEX, ADPC, ADLB, ADEFF, ADAE, ADADA) are provided as File S1 (`synthetic_data.zip`). Model versions were pinned at evaluation time. Full reproduction requires access to the specific agentic platforms (Claude Code, OpenAI Codex) and underlying LLMs (Opus 4.6, GPT-5.3), which are commercial services.

Supplementary Materials

Supporting Information S1 provides the complete task library (36 task definitions with domain rules and verification criteria), workflow protocol, outer loop pseudocode, and implementation guide. S3 contains the workflow protocol. S5 fully specifies the synthetic vorlizumab scenario including ground-truth PK/PD parameters and study designs. File S1 (`synthetic_data.zip`) provides ADaM-compliant synthetic datasets (ADSL, ADEX, ADPC, ADLB, ADEFF, ADAE, ADADA) for both studies, enabling direct reproduction of the modeling and simulation steps.

Future Directions

Community contribution is the intended growth mechanism: task definitions add capabilities, reference examples improve success rates, and benchmark scenarios enable quantitative evalu-

ation. Priority areas include additional therapeutic areas (small molecules, biologics with target-mediated disposition), pediatric and special population tasks, and integration with electronic data capture systems. Extending review tasks to present formal decision trees with quantitative evidence per branch would formalize the variation-as-exploration approach. Grounding task rules in published regulatory guidance (FDA population PK and exposure-response guidance) would reduce dependence on individual expert conventions. The framework could also be extended to clinical pharmacology reporting, bioequivalence analysis, PBPK, and QSP.

Expert Commentary

This paper makes a valuable contribution by shifting the focus from raw LLM capability to context engineering as the primary determinant of reliability in AI-assisted pharmacometrics. The divergence-driven refinement method is particularly clever—using run-to-run nondeterminism across platforms as a diagnostic to locate underspecified rules, rather than treating nondeterminism as a nuisance. The honest treatment of remaining divergence (400–600 mg dose range) as genuine methodological latitude, with parallel runs mapping the analytical decision space, is scientifically mature and more useful than collapsing to a single path. The main caveats are the small evaluation scale (single synthetic scenario, 3 replicates per platform) and the confounded harness/LLM effects, which limit the strength of cross-platform conclusions. The framework’s emphasis on traceability, file-based audit trails, and human checkpoints aligns well with emerging regulatory expectations for AI in drug development. A key open question is whether the authoring burden of maintaining a 36-task library with point-of-use rules scales to routine use across diverse therapeutic areas.

Bottom Line

Context engineering—not raw LLM capability—determines the reliability of AI-assisted pharmacometric analysis. Practitioners should embed domain rules and verification criteria directly into task definitions, run each task in a fresh LLM instance with file-based state transfer to avoid context degradation, and use cross-platform replication to identify rule gaps. Human review remains essential at critical decision points, and remaining run-to-run divergence should be interpreted as a signal of genuine methodological latitude rather than a failure to be eliminated.

☒ Figures

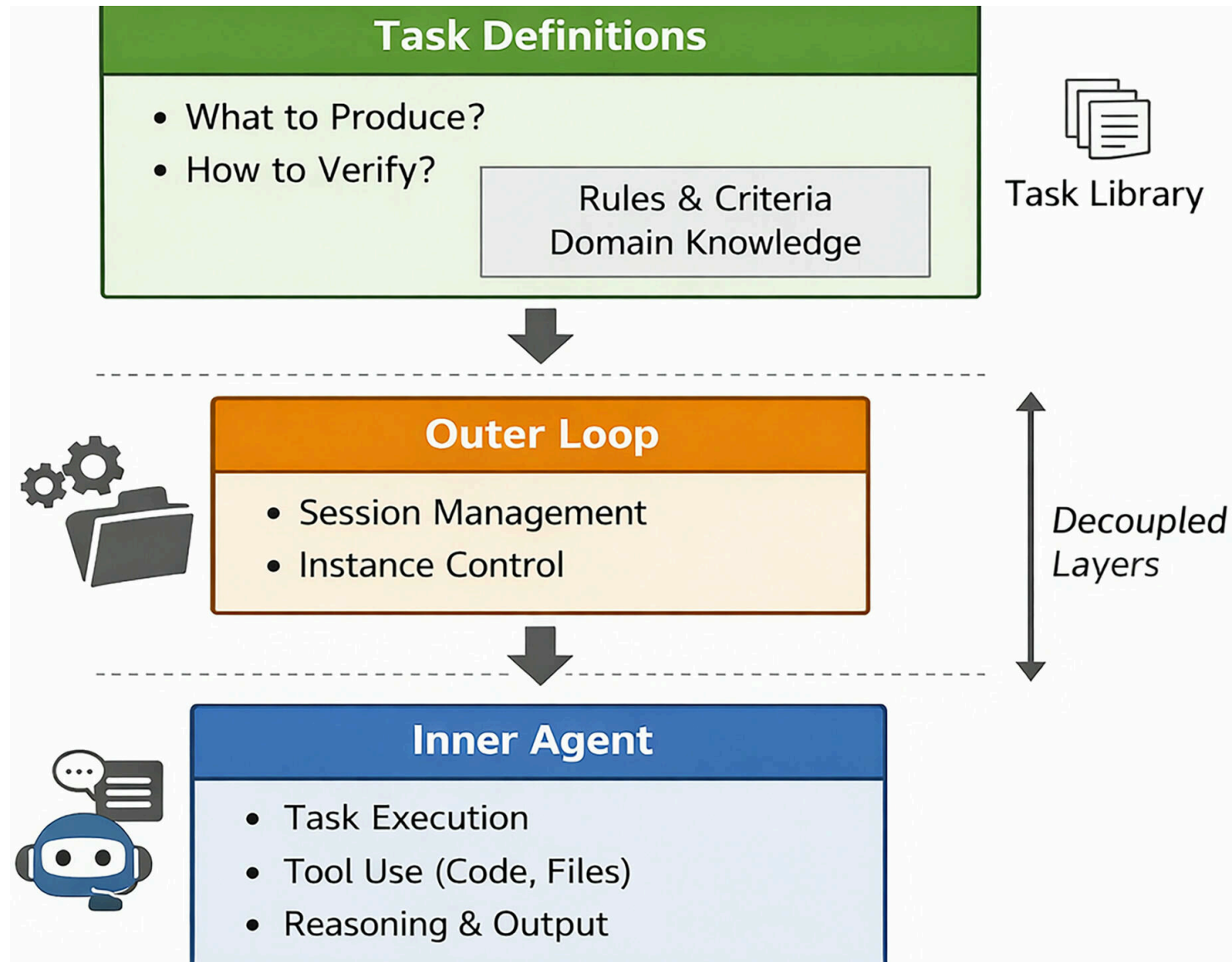


Figure 1: Three-layer architecture. Task definitions (what to produce, how to verify) are decoupled from the outer loop (instantiation strategy) and inner agent (task exec

Execution Flow

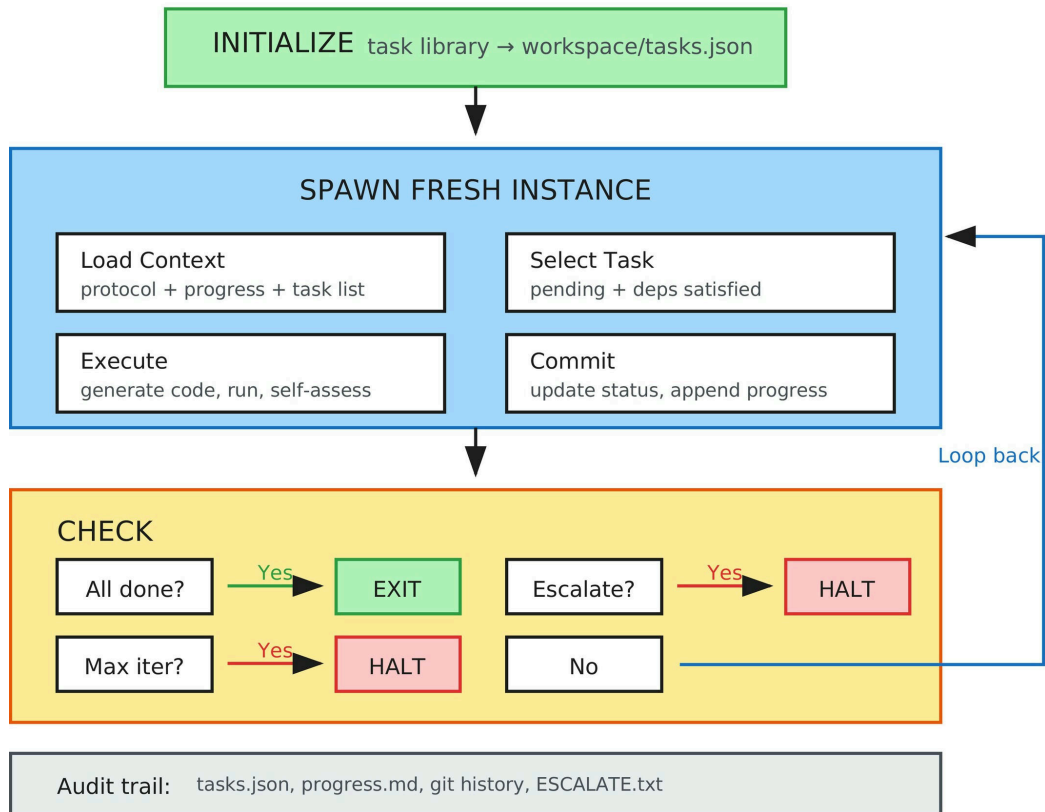


Figure 2: Execution flow. The outer loop spawns fresh agent instances using sentinel files (ANALYSIS_COMPLETE.txt,ESCALATE.txt) for control flow. Each instance selects, ex

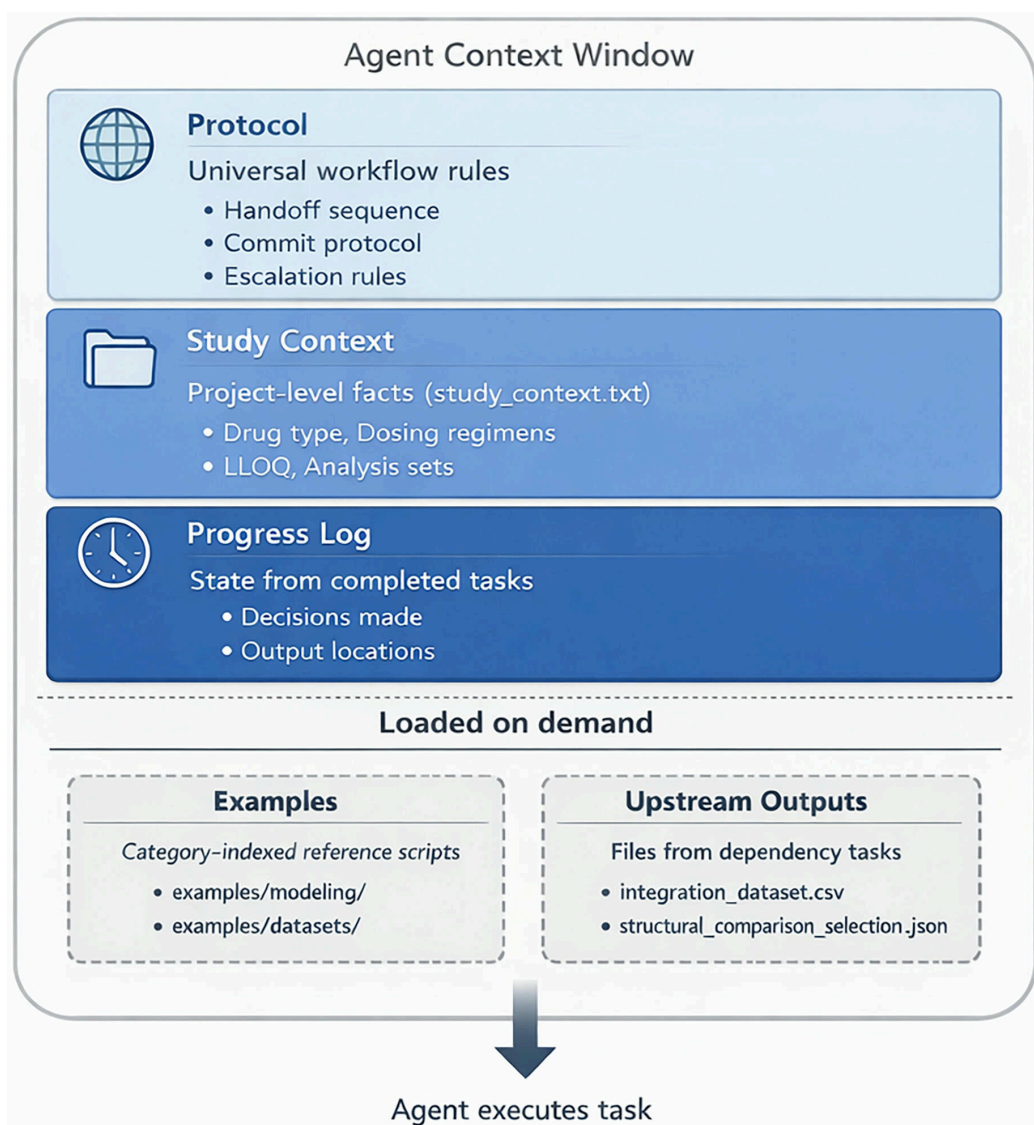


Figure 3: Context loading strategy. Solid boxes are always loaded. Dashed boxes are discoverable and loaded on demand. Study context (study_context.md) provides project-le

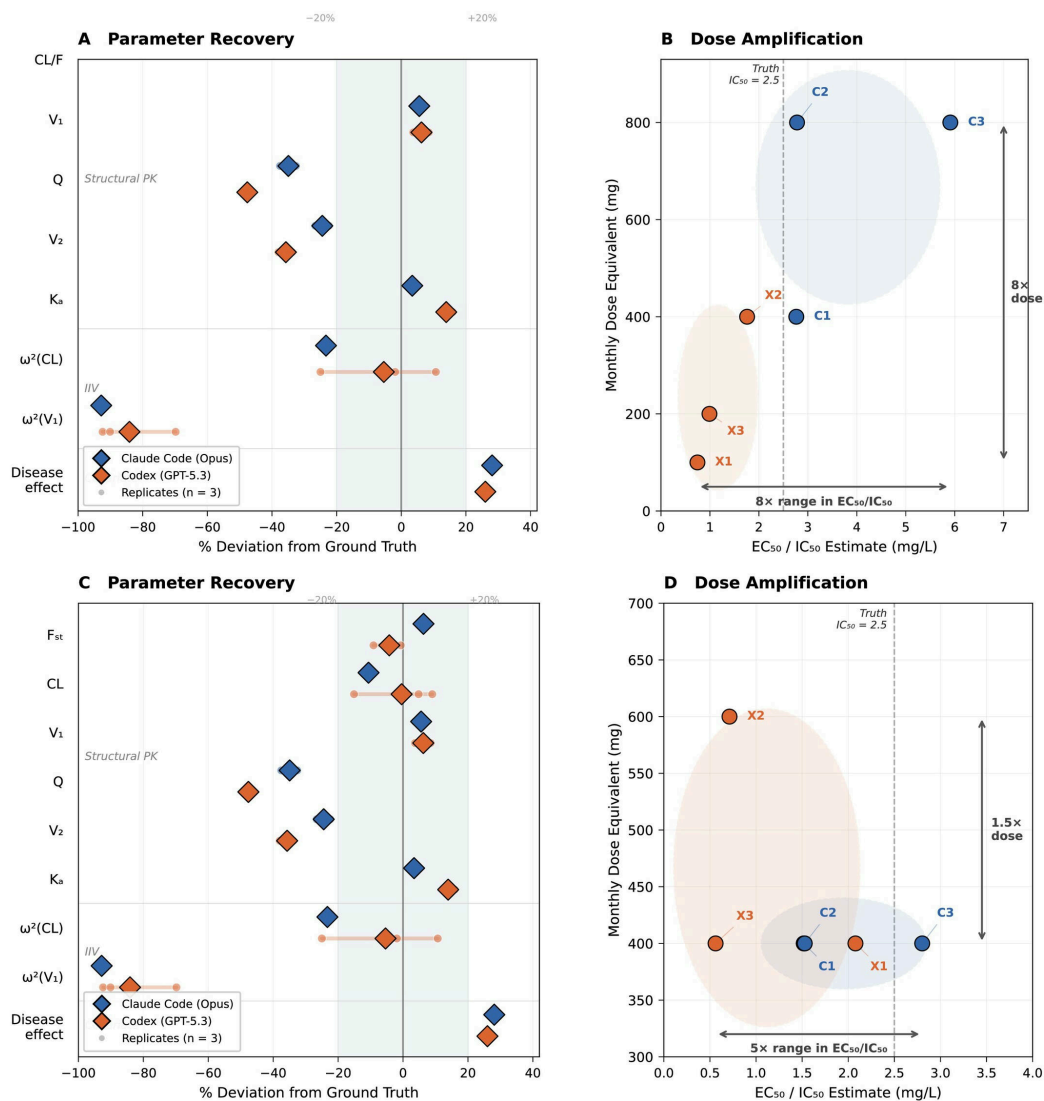


Figure 4: Cross-platform evaluation before and after rule fixes. Initial evaluation (panels A, B, top row) and post-fix re-evaluation (panels C, D, bottom row), stacked to